

Improving collaborative recommender system using hybrid clustering and optimized singular value decomposition

Zahra Movafegh^a, Abdoreza Rezapour^{b,*}

^a Department of Computer Engineering, Mehrastan University, Astaneh Ashrafieh, Guilan, Iran

^b Department of Computer Engineering, Astaneh Ashrafieh Branch, Islamic Azad University, Astaneh Ashrafieh, Guilan, Iran

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ABSTRACT

In the era of internet and big data, recommender systems are necessary to filter out useless information. Collaborative filtering (CF) is one of the most successful technique used in recommender systems, this technique suffers from a large number of users and items, or the scalability issue. In this paper, a new hybrid method based on K-means clustering (KM) and Singular Value Decomposition (SVD) which uses evolutionary algorithms is proposed to deal with scalability issue. On the one hand, KM optimized by Particle Swarm Optimization (PSO) and denoising by Density-Based Spatial Clustering of Applications with Noise (DBSCAN) is used to cluster users and reduce the number of comparisons between the target user and other users. On the other hand, SVD optimized by Genetic Algorithm (GA) is used to reduce the number of items. The proposed method is assessed on three standard datasets and the results are compared with basic CF and other extended versions that use clustering algorithms, evolutionary algorithms and dimensionality reduction techniques. The results show that our proposed method performed better than other methods in terms of *precision*, *recall* and *MAE*, and the scalability problem was improved by reducing the time complexity. Also, the combined clustering method was optimized in terms of Davies-Bouldin and the Dunn's index compared to the basic clustering methods.

1. Introduction

Today, due to the widespread use of the Internet and smart tools, a huge amount of information and resources are created on websites. In fact, we live in a world that is driven by decisions based on big data analysis (Hazem et al., 2022). “Big data” is a collection of voluminous and complex data with a certain degree of accuracy that must be properly analyzed in order to lead to the discovery of valuable information. Therefore, artificial intelligence approaches appear in various fields, including the field of information refinement and the processing of big data (Duan et al., 2019). (Q. Zhang et al., 2018). The recommender system is an example of an artificial intelligence approach to information filtering and is commonly used on e-commerce websites. This system detects the similarity in a user's preferences and tastes with other users and helps the user get closer to their destination faster among the huge amount of information available. Recommender systems have different methods, and each of these methods has its own characteristics. However, choosing the right method and using effective recommendation methods is very important for a system that provides recommendations to users (Isinkaye et al., 2015). The methods of recommender

systems are collaborative filtering, content-based filtering, knowledge-based filtering, and hybrid filtering (Geetha et al., 2018).

Among the methods mentioned, collaborative filtering is the most mature and common method that is implemented. Due to its many advantages, this method is one of the most popular among large commercial companies such as Amazon, Google, Netflix, etc. One of the main advantages of collaborative filtering compared to other methods is that it is independent of the domain. This means that this system has the ability to filter various types of items such as text, music, video, and photos (H. N. Kim et al., 2010; Patra et al., 2015). Despite its many advantages, collaborative filtering suffer from some inherent problems (Karabadi et al., 2018). Scalability is one of the problems related to this method and the reason for its occurrence is facing big data. In fact, the problem of scalability occurs due to the increase in the number of users and items.

In general, collaborative filtering systems fall into two main categories: memory-based methods and model-based methods. Memory-based methods usually use different methods of similarity measurement (Pearson, Euclidean, cosine, etc.) to find similarities between two users using the user rating matrix. High accuracy recommendation is

* Corresponding author.

E-mail addresses: Abdoreza.rezapour@iau.ac.ir, Abdoreza.rezapour@gmail.com (A. Rezapour).

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one of the important advantages of the memory-based method. In contrast, lack of scalability is one of the main problems associated with these methods.

The model-based methods use a rating matrix given by users to predict ratings. classification algorithms (Razia Sulthana and Ramasamy, 2019), clustering (Alguliyev et al., 2021), matrix dimension reduction (Ramirez-Arroyo et al., 2022), exploring association rules (Viktoratos et al., 2018), etc. are examples of model-based methods. Although these methods do not have the accuracy of memory-based methods, but in combination with memory-based methods, can solve the problem of their lack of scalability to some extent and reduce the prediction time. For this reason, in this plan, we have combined memory-based and model-based collaborative filtering methods to use in our proposed recommender system.

In this study, a new collaborative filtering method is proposed to improve the performance of the recommender system and overcome the problem of lack of scalability. This research tries to implement several different clustering methods, apply matrix dimension reduction, and finally optimize the whole proposed structure in order to better process big data, increase the accuracy of the recommendations, and help Reduction computational costs and work time. The innovation of this plan is the simultaneous use of two optimized clustering methods and optimized single eigen values, which are used to control a large number of users and items, respectively. The combination of *KM* clustering optimized by the *PSO* and *DBSCAN* clustering optimized by a *GA* to clean clusters from noise is another innovation of this project that has not been used by recommender systems so far. Using the *SVD* method and its optimization by a *GA* is one of the other innovations of this project in order to solve problems related to big data.

This work's main contribution is as follows.

- Improving the efficiency of collaborative filtering recommender systems by combining memory-based and model-based methods.
- Improving the lack of scalability of recommender systems at the user level by using clustering and at the item level by using *SVD*.
- Improving the quality of clustering by combining two methods, optimized *KM* and optimized *DBSCAN*, and removing noise.
- Optimizing the *KM* algorithm by combining it with *PSO* algorithm and *DBSCAN* by combining it with *GA*.
- increasing the quality of the *SVD* method by combining it with the *GA*.

The structure of the article is as follows: In Section 2, the research background is introduced. In Section 3, the theoretical foundations are explained. In Section 4, the proposed method is presented. In Section 5, the data set and the results obtained are presented. Section 6 contains the conclusion.

2. Research background

Different approaches have been proposed for recommender systems. Hence some of the previous achievements in various areas of this system were briefly discussed and our work were related to them.

2.1. Collaborative filtering

(Da Silva et al., 2016), proposes an evolutionary approach, called *Invenire*, to automate the choice of techniques used by combining results of different recommendation techniques. This is a new approach that uses a search algorithm to optimize the techniques combination, and can inspire hybrid methods and expert systems on how automate them. To evaluate the proposal, experiments were performed with a data set from *MovieLens* and different collaborative filtering approaches. The results obtained show that the *Invenire* outperforms all collaborative filtering approach separately in all contexts addressed. Thus, the proposal was able to increase the accuracy on the generated

recommendations and automate the combinations of techniques.

(Fong et al., 2008), presented a *GA*-based approach for supporting combined modes of collaborative filtering. *GA*-based *CF* provides reasonably good recommendation accuracy. In particular, they show that how the input variables can be coded into *GA* chromosomes in various modes. They improved performance by incorporating user profile features in chromosomes. They conducted their experiments using different sets of feature on the *MovieLens* and *IMDB* data sets to show how and which feature weights affect fitness and performance.

(Katarya, 2018), proposed a hybrid recommender system which utilized *k-means* clustering algorithm with bio-inspired artificial bee colony (*ABC*) optimization technique and applied to the *MovieLens* data set. The proposed system (*ABC-KM*) is compared with existing approaches, and the consequences have been examined. Estimation procedures such as precision, mean absolute error, recall, and accuracy for the movie recommender system delivered improved results for *ABC-KM* collaborative movie recommender system. The experiment outcomes on *MovieLens* data set established that the projected system provides immense achievement regarding scalability, performance and delivers accurate personalized movie recommendations by reducing cold start problem.

(Kilani et al., 2018), build a novel genetic-based *CF RS* that hybridizes both the neighborhood and the latent factor models to predict items for the active user. The main difference between their *RS* and conventional *RS* is that, they only consider the users and items that are related to the active user rather than considering the whole users and items. Using three different data sets: *MovieLens*, *FilmTrust*, and *CiaoDVD*, they experimentally show that their model improved in term of *MAE*, *precision*, and *recall*.

(Ahmed et al., 2022), in their *RS* is considering a few factors such as buying records, classification or clustering items, and user's geographic location. Collaborative filtering algorithms (*CFAs*) are much more common approaches for cooperating to mesh the respective documents they retrieved from the historical data. *CFAs* are distinguished in plenty of features that are uncommon from other algorithms. In this existing system classification, precision and efficiency and error rate are statistical measurements that need to be enhanced according to the current need to fit for global requirements. The proposed work deals with enhancing accuracy levels of text reviews with the recommender system while interacting by the numerous users for their domains. The authors implemented the recommender system using a user-based *CF* method and presented the significance of collaborative filtering on the movie domain with a recommender system. This whole experiment has been implanted using the *RapidMiner* Java-based tool. Results have been compared with existing algorithms to differentiate the efficiency of the current proposed approach.

(Do and Nguyen, 2022; Khurana and Dhingra, 2021) propose novel semantic-enhanced *NCF* models which are applied to movie rating prediction and movie recommendation. Therefore, *MovieLens* and *IMDB* data sets are taken into account as case studies. The proposed models are the integration of ontology-like modeling and deep learning for recommendation tasks into two parts: (1) building the semantic knowledge base for movies and (2) building the user behavior analytic model that has semantic knowledge inference on the knowledge base combined with the sequential preference learned from user sessions, input into the *NCF* module for making predictions or recommendations. Several experiments have been conducted to show their better recommendation performance than the traditional *NCF* model.

(Nassar et al., 2020), proposed a novel multi-criteria collaborative filtering model based on deep learning. Their model contains two parts: in the first part, the model obtains the users and items' features and uses them as an input to the criteria ratings deep neural network, which predicts the criteria ratings. Those criteria ratings constitute the input to the second part, which is the overall rating deep neural network and is used to predict the overall rating. Experiments on a real-world data set demonstrate that their proposed model outperformed the other

state-of-the-art methods, and this provides evidence pointing to the success of employing deep learning and multi-criteria in recommendation systems.

2.2. Content-based filtering

(Kant and Bharadwaj, 2012), attempted towards developing effective content based filtering (CBF) by introducing an item representation scheme, fuzzy similarity measures and incorporating collaborative diverse predictions for alleviating its recommendation diversity. The proposed work deals with the use of fuzzy modeling for the representation of items and user behavior model and also addresses the recommendation diversity problem of *CB-RSs*. Experimental results show that the proposed hybrid scheme *Fuzzy-CF-CBF* outperforms hybrid *CF-CBF*, as well as both the fuzzy collaborative filtering (*Fuzzy-CF*) and the fuzzy content based filtering (*Fuzzy-CBF*).

(Son and Kim, 2017), propose a novel *CBF* method that uses a multiattribute network to effectively reflect several attributes when calculating correlations to recommend items to users. In the network analysis, they measure the similarities between directly and indirectly linked items. Moreover, their proposed method employs centrality and clustering techniques to consider the mutual relationships among items, as well as determine the structural patterns of these interactions. the proposed approach was compared with existing approaches using MovieLens data, their approach outperformed existing methods in terms of accuracy and robustness. Their proposed method can address the sparsity problem and over specialization problem that frequently affect recommender systems. Furthermore, the proposed method depends only on ratings data obtained from a user's own past information, and so it is not affected by the cold start problem.

(D. Wang et al., 2018), present a system that recommends suitable journals or conferences with a priority order based on the abstract of a manuscript. To follow the fast development of computer science and technology, a web crawler is employed to continuously update the training set and the learning model. To achieve interactive online response, they propose an efficient hybrid model based on chi-square feature selection and softmax regression. Their test results show that, the system can achieve an accuracy of 61.37% and suggest the best journals or conferences in about 5s on average.

2.3. Knowledge-based filtering

(Cena et al., 2021), Stated that the aim of the paper is to provide logical foundations for knowledge-based recommender systems, for which, unlike other problem solving tasks, a comprehensive formalization is not yet available. This aim is justified by the need to compare recommenders based on the way they use knowledge to generate recommendations and, consequently, on the underlying semantics of the recommendation process itself. Moreover, since the here adopted logical formalization has been borrowed from other tasks such as diagnosis, many interesting results and opportunities can be transposed from such tasks to recommendation. While they do not aim at proposing a new recommendation generation technique, they believe that their formalization will be the basis for unifying different approaches to knowledge-based recommendation, revealing their semantics and offering a conceptual framework to compare them. In fact, the framework covers different variations of knowledge-based recommendation, such as context-aware, constraints-based, package and group recommendation, as well as recommendation based on negative preferences.

(Dong et al., 2020), propose an interactive, personalized design knowledge-based recommender system. This system works through a newly proposed design process: consumers' emotional requirement identification, design schemes generation, recommender, 3D virtual prototype display and evaluation, design factors adjustment. This process can be performed repeatedly until the designer is satisfied. The proposed system has been validated through a number of successful real

design cases.

(Tarus et al., 2017), proposed a hybrid knowledge-based recommender system based on ontology and sequential pattern mining (*SPM*) for recommendation of e-learning resources to learners. In the proposed recommendation approach, ontology is used to model and represent the domain knowledge about the learner and learning resources whereas *SPM* algorithm discovers the learners' sequential learning patterns. Their approach involves four steps: (1) creating ontology to represent knowledge about the learner and learning resources, (2) computing ratings similarity based on ontology domain knowledge and making predictions for the target learner, (3) generation of top N learning items by the collaborative filtering recommendation engine, and (4) application of *SPM* algorithm to the top N learning items to generate the final recommendations for the target learner. A number of experiments were carried out to evaluate the proposed hybrid recommender system and results show improved performance. Furthermore, the proposed hybrid approach can alleviate both the cold-start and data sparsity problems by making use of ontological domain knowledge and learner's sequential access pattern respectively before the initial data to work on is available in the recommender system.

2.4. Hybrid filtering

(Nilashi et al., 2018), develops a new hybrid recommendation method based on Collaborative Filtering (*CF*) approaches. in this research, they solve two main drawbacks of recommender systems, sparsity and scalability, using dimensionality reduction and ontology techniques. Then, they use ontology to improve the accuracy of recommendations in *CF* part. In the *CF* part, they also use a dimensionality reduction technique, Singular Value Decomposition (*SVD*), to find the most similar items and users in each cluster of items and users which can significantly improve the scalability of the recommendation method. they evaluate the method on two real-world data sets to show its effectiveness. The results showed that their method is effective in improving the sparsity and scalability problems in *CF*.

(Mohammadpour et al., 2019), proposed a single-objective hybrid evolutionary approach for clustering items in the offline collaborative filtering *RS*. This method, after generating a population of randomized solutions, at each iteration, improves the population of solutions first by Genetic Algorithm (*GA*) and then by using the Gravitational Emulation Local Search (*GELS*) algorithm. Simulation results on standard data sets indicate that although the proposed hybrid meta-heuristic algorithm requires a relatively high run time, it can lead to more appropriate clustering of existing data and thus improvement of the Mean Absolute Error (*MAE*), Root Mean Square Error (*RMSE*) and Coverage criteria.

(Salehi et al., 2013), proposes a hybrid recommender system for learning materials based on their attributes to improve the accuracy and quality of recommendation. The presented system has two main modules: explicit attribute-based recommender and implicit attribute-based recommender. In the implicit attribute-based module, genetic algorithm is used for extracting implicit attributes of learners from historical rating in the shape of weight vectors. Then, recommendation is generated by Nearest Neighborhood Algorithm (*NNA*). In the explicit attribute-based module, preference matrix (*PM*) is introduced that can model the interest of learners based on explicit attributes of learning materials in a multidimensional space. Then, a new similarity measure between *PMs* is introduced and recommendations are generated by *NNA*. The experimental results show that their proposed method outperforms current algorithms on accuracy measures and can alleviate some problems such as cold-start and sparsity.

(Paradarami et al., 2017), present a deep learning neural network framework that utilizes reviews in addition to content-based features to generate model based predictions for the business-user combinations. They show that a set of content and collaborative features allows for the development of a neural network model with the aim of minimizing logloss and rating misclassification error using stochastic gradient

Table 1

List of symbols and notations.

Notation	Description
a	Target user.
u	User.
$W_{a,u}$	similarity of the target user a with the user u .
$r_{a,i}$	rating the target user a to the item i .
$r_{u,i}$	rating user u to the item i .
$\bar{r}_a$	average of all ratings of the target user a .
$\bar{r}_u$	average of all ratings of the user u .
$P_{a,i}$	predicts rating the target user a to item i .
h	nearest neighbors to the target user.
k	the number of cluster for KM .
ϵ	maximum neighborhood radius.
mpt	minimum number of points in the ϵ neighborhood.
p_{id}	the i th candidate particle in a d dimensional space.
S_{id}	the i th candidate chromosome in a d dimensional space.
c_{ij}	the center of the j th cluster in the i th particle.
$p_{i \text{ best}}$	the best position of the particle.
$p_{g \text{ best}}$	the best position of the particle found among the total particles.
W	coefficient of inertia.
c_1	constant coefficients for the $p_{i \text{ best}}$.
c_2	constant coefficients for the $p_{g \text{ best}}$.
λ	Eigen value.
τ	Threshold for Dimension reduction in SVD

descent optimization algorithm. They perform computational experiments using the Yelp Academic Data set and show that the hybrid approach is a very promising solution when compared to standalone memory-based collaborative filtering method.

(Khurana and Dhingra, 2021), proposed an improved hybrid and knowledge based recommender system by integrating the clustering method within content based filter and classification method within collaborative filter. The proposed method handled the scalability problem by using the fuzzy clustering method. This reduced dimension based data set is processed by the probabilistic Bayesian network classifier for predicting the recommendations. The sparsity problem is handled in both stage of this model. The proposed recommender system model is applied on MovieLens data set. The comparative analysis was done against content-based recommender system (CBRS), Pearson correlation based collaborative recommender system (PCRS), Frequency-weighted Pearson Correlation (FPC), Weighted Pearson Correlation (WPC) and hybrid recommender systems (HRS). The significant reduction in MAE and RMSE rate is identified in this work. The experimentation results identified that the proposed model reduced the error rate and improved the accuracy rate over existing systems.

3. Theoretical foundations

In the following sections, related topics for the development of the proposed recommender system and a brief description of them are provided. complete list of symbols used are provided in Table 1.

3.1. Recommendation system based on collaborative filtering

Collaborative filtering-based recommender systems use comments and ratings given by users to provide recommenders. This is done by searching a large set of users and creating a smaller set of users with similar tastes to the target user.

Suppose there is a list of users and items including $U = \{u_1, u_2, u_3, \dots\}$ and $I = \{i_1, i_2, i_3, \dots\}$. As it can be seen in Table 2, users can rate their favorite items in certain ranges.

Collaborative filtering systems go through three main steps to provide advice to the target user (Da Silva et al., 2016).

- The first step in a collaborative filtering system is to find similarities between users. Equation (1) is called the Euclidean distance method

and is one of the methods for measuring similarity (based on distance) between users.

$$W_{a,u} = \sqrt{\sum_{i=1}^m (r_{a,i} - r_{u,i})^2} \quad (1)$$

$W_{a,u}$ is the similarity of the target user a with the user u , $r_{a,i}$ is the rating the target user a to the item i , and $r_{u,i}$ is the rating user u to the item i .

- The second step is to select a subset of more similar users.
- The third step is to predict the rating of unrated items. The prediction calculation is performed according to Equation (2).

$$P_{a,i} = \bar{r}_a + \frac{\sum_{u=1}^h (r_{u,i} - \bar{r}_u) * W_{a,u}}{\sum_{u=1}^h |W_{a,u}|} \quad (2)$$

In this equation $P_{a,i}$ predicts rating the target user a to item i . $\bar{r}_a$ is the average of all ratings of the target user a . $\bar{r}_u$ is the average of all ratings of the user u . h represents the nearest neighbors to the target user. Finally, the output of the collaborative filtering recommender system is the best items recommended to the target user.

3.2. Clustering

Clustering refers to placing data in several groups, so that the data of one cluster are most similar to each other but have the least possible similarity with the data in other clusters (Jaaz et al., 2022). Clustering is a common technique for data analysis in many fields, including pattern recognition, image analysis, data compression, information retrieval, etc. (Ghosal et al., 2020). There are different types of clustering algorithms with different functions. This paper will use the combination of KM and $DBSCAN$ algorithms for clustering.

3.2.1. K-means (KM) clustering algorithm

KM is one of the simplest and most popular unsupervised learning algorithms (K. jae Kim and Ahn, 2008). This algorithm is a partitional clustering method and divides the data into k separate clusters based on their distance measure, so that the squared error between a cluster and the points of that cluster is minimized (Prakash K et al., 2022). Suppose $O = \{o_1, o_2, \dots, o_n\}$ is a set of n data samples that must be clustered into a set of k clusters, $C = \{c_i, i = 1, \dots, k\}$. The aim of KM clustering is to minimize the sum of squared error over all k clusters and it is defined in equation (3):

$$J(c) = \sum_{i=1}^k \sum_{o_j \in c_i} (o_j - z_i)^2 \quad (3)$$

In this equation, o_i, z_i, c_i and k respectively represent the i th cluster, the center for the i th cluster, the data sample belonging to the i th cluster and the total number of clusters.

KM can be widely used when dealing with big data (Capó et al., 2017), medical data (Khanmohammadi et al., 2017), industrial data (Mezghani et al., 2018), film data (Cai and Wang, 2020), etc. However, this algorithm, despite its many advantages, may be affected by noisy data.

3.2.2. Density-Based Spatial Clustering of Applications with noise (DBSCAN)

$DBSCAN$ is a common density-based clustering method (Latifi-Pakdehi and Daneshpour, 2021). In this algorithm, a cluster is created in any part of the space where the data is denser. Unlike many clustering methods, $DBSCAN$ is not sensitive to the shape of the data and can also detect noise points and not insert them into any of the clusters. In fact, $DBSCAN$ is the most widely used method to remove noise from data. For

example, this algorithm can be used to remove data noise related to intrusion detection systems (Saxena et al., 2019), machine learning (Shi et al., 2019), image processing (L. Zhang et al., 2022), etc. DBSCAN uses two parameters, ε and mpt , to determine the density and perform clustering.

ε : maximum neighborhood radius.

mpt : the minimum number of points in the ε neighborhood.

Some central concepts related to the DBSCAN are reviewed as follows (Zhu et al., 2021).

Definition 1. (ε neighborhood): For the ε neighborhood of a randomly selected object p in a data set D , which is often determined by a distance function, and can be defined by: $\varepsilon(p) = \{q \in D | \text{dist}(p, q) \leq \varepsilon\}$.

Definition 2. (Directly density-reachable): An object q is directly density-reachable from an object p if two conditions are satisfied: (a) $q \in \varepsilon(p)$; (b) $|\varepsilon(p)| \geq mpt$.

Definition 3. (Core object & border object): A data point can be regarded as the core object if it satisfies the condition of definition 2. Whereas the border object is not a core object itself but can be density-reachable from other core objects (see definition 4).

Definition 4. (Density-reachable): Under the condition of ε and mpt , an object q is density-reachable from an object p if there is a chain of points: $p_1, \dots, p_n$ between them, meanwhile $p_1 = p, p_n = q$, and any object p_{i+1} is directly density-reachable from p_i .

Definition 5. (Density-connected): If there exists an object $o \in D$ such that both p and q are density-reachable from o . In this case, the objects p and q are mutually density-connected.

Definition 6. (Cluster): For the data set D , a subset C under the conditions of ε and mpt can be as a cluster needs to satisfy: (1) $\forall p, q$, if $p \in C$ and q is density-reachable from p , then $q \in C$. (2) $\forall p, q \in C$, p and q are directly density-reachable or density-connected.

Definition 7. (Noise): The noise is a set of objects in the data set D , which not belonging to any cluster since it cannot satisfy ε and mpt , i.e. noise = $\{p \in D | \forall i : p \notin c_i\}$ where $i = 1, 2, \dots, k$.

3.3. Particle Swarm Optimization (PSO) algorithm

PSO is a meta-heuristic algorithm based on swarming intelligence and inspired by different biological species, such as the mass flight of birds or the mass movement of fish (Bakshi et al., 2014). PSO is initialized with a number of solutions or in other words particles. Each particle moves in the search space look for the best local position and the best global position. In fact, the most effective search strategy to find the best particle is to know which particle is nearest to the optimal position and follow it. PSO must have a fitness evaluation function that takes the particle's position and assigns to it a fitness value. The aim is to optimize the fitness function. Also, each particle has its own velocity and position to change the flight direction in the predefined search space. The velocity of each particle and the position of each particle are mentioned in equations (8) and (9), respectively (refer to section 4-2). All particles move in the search space following the current optimal particles until reach the stopping condition (Chiu et al., 2009).

How the particles move from the current position to the new position is shown in Fig. 1.

PSO is used in various fields such as: medical sciences, communication networks, clustering and classification, combined optimization, prediction, neural networks, etc. (Poli, 2008).

3.4. Genetic algorithm

A GA is an optimization method that imitates nature and biological

methods to optimize problems (Da Silva et al., 2016). The initial population in the GA is affected by three main operators (selection operator, crossover operation and mutation operator) and simulates the process of natural evolution. After applying the operators, a new population of individuals is generated in the solution space of the problem. This method is repeated until the stopping criterion is reached. In general, this algorithm is a random search method and is efficient at solving problems with a large search space.

GA has proven useful in solving various problems in fields such as artificial intelligence, engineering, medical sciences, economics, physics, chemistry, etc. (Garai, 2022).

In this paper, the GA is used to optimize clustering algorithms and single value decomposition (Kapil et al., 2016; Yao et al., 2022).

Genetic algorithm operators (Xue et al., 2021) are summarized as follows.

Selection Operator

This operator selects the best individuals in the population as parents to produce the next generation. The most common method used for selection is selection based on the roulette wheel. The idea of this method is probabilistic selection. Therefore, in this paper the selection probability of each chromosome based on its cost function is obtained through Equation (4).

$$Probability_i = \frac{chromosome\ cost_i}{\sum_j chromosome\ cost_j} \quad (4)$$

$$chromosome\ cost_i = e^{\left(-\beta * C_i / C_{max}\right)} \quad (5)$$

Crossover Operator

This operator is the main method for generating new people. In fact, crossover with the combination of parents' GA information leads to the production of children.

Mutation Operator

By making changes in the genes of the children, the mutation operator provides the ability to maintain population diversity and increases the possibility of achieving global optimization.

3.5. Singular value decomposition (SVD) method

SVD is one of the powerful data reduction techniques and a powerful computational tool for data analysis in linear algebra (Nilashi et al., 2018). SVD is used in various applications, such as: solving homogeneous linear equations, approximation of a low rank matrix, nearest orthogonal matrix, range and null space of a matrix, etc. In addition, SVD has interesting applications in machine learning and image processing (Guo et al., 2016). Also, SVD is often used in matrix factorization in recommender systems. This parsing method can combine the relationships between users and items by some factors linearly. As shown in Fig. 2, SVD converts the ranking matrix A with dimensions $m \times n$ into three low-degree matrices U and Σ and V as $A_{mn} = U_{mm} \Sigma_{mn} V_{nn}^t$ (Walek and Fojtik, 2020). Here, A represents the user's rating matrix for items. U refers to the matrix of user concepts. Σ represents single values and V^t is a rewrite of the concepts of the items.

One of the problems associated with SVD is that this algorithm does not know how many elements of the original diameter to consider as a threshold (a threshold provides information about how much dimension should be reduced). Therefore, in most cases, it is enough to choose a number of larger elements. Nevertheless, we intend to choose the appropriate and optimal threshold value with the help of GA.

4. The proposed method

In general, collaborative filtering recommender systems face large volumes of data and therefore suffer from a number of problems, including lack of scalability. The use of clustering and matrix dimension reduction methods can greatly reduce this problem and have a positive effect on the accuracy and speed of the recommender system. In this article, we use the *KM* algorithm to perform clustering. *KM* is one of the simple and effective methods of clustering. But this algorithm suffers from two basic problems. First, its accuracy depends entirely on the primary selection centers, and second, it is affected by the presence of noise (Yu et al., 2018). To solve first problem, we combine *KM* with *PSO* algorithm and solve the second problem by combining optimized *KM* with optimized *DBSCAN* by *GA* algorithm. we call this combined clustering *PSO-KD*. Then, on each of the clusters, we perform dimension reduction operation by *SVD* optimized by *GA*. Our aim is to increase the accuracy of the recommender system, reduce the execution time, reduce the computational costs and finally provide the best recommendation list to the target user.

Algorithm 1. The Main Procedure of Proposed Method

INPUT: user-item matrix

OUTPUT: Recommendation list

Step1: Preprocessing and dividing data.

Arrange and clean the rating matrix.

Split the rating matrix into test and train by cross validation.

Hide some test data using ALLBUT10, ALLBUT5, GIVEN15, and GIVEN10 strategies.

Step 2: Clustering by *KM* and *PSO*.

Create initial particles.

/*as seen in figure 3 (particle-length=k*n, k=no-clusters, n=no-items */

for t=1: Max_Gen

Run *KM* and *PSO* // figure 5

Step 3: Cleaning *KM* clusters by *DBSCAN* and *GA*.

Create initial chromosomes.

/*as seen in figure 4 (chromosome-length=3, in this step we need the first 2 genes i.e. ϵ and mpt)*/

for t=1: Max_Gen

for c=1: k (no-clusters)

Run *DBSCAN* and *GA* // Algorithm 2

end for

end for

step 4: matrix dimension reduction by *SVD* and *GA*.

/* in this step we need the gene 3 i.e. τ */

for t=1: maximum generation

for c=1: k (no-clusters)

Run *SVD* and *GA* // Algorithm 2

end for

end for

/*It should be noted that *SVD* can be run after the *DBSCAN* in the loops of step 3*/

Step 5: Recommendation.

Assign the active user to the most similar cluster.

Calculate similarity and Select the most similar users to active user // Equation 1

Provide a list of top recommendations to the active user.

4.1. Data processing

First, by performing processing on the data of recommender systems, we convert the data into a clean rating matrix containing the users' ratings to items. Then, in order to performance the proposed model, we convert the rating matrix into two sets of train and test. And we perform the model on the train data set.

4.2. Clustering by *KM* + *PSO* method

In this section, an improved clustering algorithm based on a combination of *KM* and *PSO* algorithms is first used to determine the vector of cluster centers. In addition to maintaining the quality of *KM*, this combination also solves its convergence problem and has very powerful capabilities in multidimensional space. Each particle contains the vector of the centers of the existing clusters. In fact, a group of particles $P = \{p_{1d}, p_{2d}, \dots, p_{id}\}$ were considered in a continuous solution space. The particle structure can be seen in Fig. 3. p_{id} represents the i th candidate particle for the optimization problem in a d dimensional space. c_{ij} represents the center of the j th cluster in the i th particle. In each particle, the structure of each c_i cell is also shown as $c_i = (r_{i1}, r_{i2}, \dots, r_{in})$. In this

structure, n represents the number of available items and r_i also indicates the rating given to the items.

The clustering responses in our method are performed by minimizing the objective function. Therefore, quantization error were used to measure particle fit (Chiu et al., 2009).

$$J_e = \frac{\sum_{j=1}^k \left[\sum_{x \in n_{ij}} d(x, c_{ij}) \right] / |n_{ij}|}{k} \quad (6)$$

$$d(x, c_{ij}) = \|x - c_{ij}\| \quad (7)$$

Here: d is defined in Equation (7), $\|x - c_{ij}\|$ represents the distance of the data from the center of the clusters. $|n_{ij}|$ is the number of data vectors belongs to the n_{ij} cluster, ie the frequency of that cluster.

The combination of *KM* and *PSO* is as follows.

- 1) Select the k number of centers of the cluster at random.
- 2) Calculate the Euclidean distance $d(x, c_{ij})$ to all centers of the n_{ij} cluster.
- 3) Assign x to the n_{ij} cluster, such that $d(x, c_{ij}) = \min_{j=1, \dots, k} \{d(x, c_{ij})\}$.
- 4) Calculate the fit using Equation (6).
- 5) Inject the resulting solution of *KM* as one of the primary particles to the optimal particle swarm.

INPUT: Range of each gens in chromosome (Table 10) and GA parameters (Table 9)

OUTPUT: Optimized SVD and DBSCAN

```

1: Generate populations (pop-size) // Chromosomes ( $\epsilon$ ,  $mpt$ ,  $\tau$ )
2: for t=1: Max-G
3:     for p=1: pop-size
4:         for c=1: k (no-clusters)
5:             DBSCAN ( $\epsilon$ ,  $mpt$ );
6:             SVD ( $\tau$ );
7:         end for
8:         Cost-Function (); // Equation 12
9:     end for
10:    Select (); // Roulette wheel
11:    Crossover (); // Two point
12:    Mutation ();
13: end for
14: return Optimal Solution (Genetically Optimized SVD and DBSCAN)

```

- 6) Record the best personal and global situations.
- 7) Update the centers of the clusters using Equations (8) and (9).

$$v_i(t) = W * v_i(t-1) + c_1 * rand_1 * (p_{i \text{ best}} - x_i(t-1)) + c_2 * rand_2 (p_{g \text{ best}} - x_i(t-1)) \quad (8)$$

$$x_i = x_i(t-1) + v_i(t) \quad (9)$$

Equation (8) corresponds to a new particle velocity in which: W is the coefficient of inertia. c_1 and c_2 are constant coefficients for the best particle position and the best particle position found among all particles, respectively. $rand_1$ and $rand_2$ are two random numbers with uniform distribution in the range $[0,1]$. $p_{i \text{ best}}$ and $p_{g \text{ best}}$ are the best positions of the particle and the particle found among the total particles, respectively. Equation (9) is related to the new position of the particle and is obtained from the sum of the old position with the new velocity of the particle.

4.3. Apply genetic algorithm

In this section, the parameters in the proposed method are optimized by a GA. The parameters are: ϵ and mpt in the DBSCAN algorithm as well as τ to reduce the dimensions in the SVD method. In order to optimally select values by GA, a lower bound and an upper bound for each parameter is set. We take an experimental value for ϵ and mpt , then select the lower and upper bounds around the desired value. To find the τ boundaries, we use Equation (10) and consider the maximum λ as the upper bound.

$$0 \leq \lambda \leq \sum_{j=1, j \neq i}^n |u_{ij}| + u_{ii} \quad (10)$$

In this equation: λ is the eigen value in SVD. u_{ii} represents user u in the main diameter of the matrix, and u_{ij} represents user u in row i and column j .

Each chromosome carries the listed parameters. The structure of the chromosomes can be seen in Fig. 4. In this structure, S_{id} represents the i solution to the optimization problem. The range of each gene is $\epsilon = [lb, ub]$, $mpt = [lb, ub]$, $\tau = [lb, ub]$. lb and ub represent the lower and upper bounds, respectively.

Algorithm 2. GA + DBSCAN and GA + SVD

4.3.1. Cleaning KM clusters by GDBSCAN

Up to this part, initial clustering was performed with optimized *KM*. However, clustering performed by *KM* is sensitive to outlier data or noise (in data analysis, an outlier is an observed value that is far from

Table 2
User rating matrix.

	i_1	i_2	i_3	i_4	i_5
u_1	3	1	–	5	–
u_2	2	1	4	5	5
u_3	–	5	–	–	1
u_4	–	2	3	–	–
u_5	1	–	4	2	–

other values). To solve this problem, we use *DBSCAN* optimized by *GA* (*GDBSCAN*) and apply it on each of the clusters generated by *KM + PSO*. After the denoising operation, the proposed hybrid clustering process (*PSO-KD*) is finished and we have optimal and clean clusters. The steps of *PSO-KD* are given in Fig. 5.

4.3.2. Matrix dimension reduction by GSVD

After performing the clustering hierarchy, we want to do the prediction work and, based on that, provide recommendations to the target user. To do this, the target user is assigned to one of the clusters. Now, to make a prediction, the similarity of the target user with all users in the cluster must be calculated. Doing so in high-dimensional problems leads to high computational costs as well as method weaknesses. To deal with this problem, we use the matrix dimension reduction method. The

elements, a number of larger elements are sufficient. Determining these larger elements is very important and can affect the end result. Therefore, we will consider a threshold value using the *GA* and keep the values larger than the threshold in the matrix and remove the rest.

4.3.3. Cost function of genetic algorithm

The *GA* can calculate the accumulated error for each solution in a given population. Here, the accumulated values of the square root of the error squares are used to form the cost function. Algorithm 3 shows how the root mean square error is calculated by the *GA* for each of the solutions and for the Euclidean technique used in the problem.

Algorithm 3. Calculating the RMSE

Input: Data Set
Output: RMSE

```

1: Select  $M_t$  from Data set //test matrix
2:   for each user in  $M_t$ 
3:      $RMSE_a = 0$ ;
4:     Generate  $M_a$  //remove k ratings for each test user
5:     for each removed ratings in  $M_a$ 
6:        $RMSE_k = 0$ 
7:       Predict removed ratings
8:       Compute  $RMSE_k$ 
9:        $RMSE_a = RMSE_a + RMSE_k$ 
10:    end
11:     $RMSE_{tot} = RMSE_{tot} + RMSE_a$ 
12:  end

```

method used is *SVD* optimized by *GA* (*GSVD*). Due to the nature of *SVD* and its linearity, we apply it to the matrix and achieve three lower degree matrices. For greater simplicity and efficiency, using the *GA*, we consider a threshold for the parsed matrix, and to calculate the similarity of the new user with other users, we use only the dimensions that are selected before the threshold. It should be noted that as the number of dimensions of the problem decreases, the dimensions of the new user should also decrease accordingly.

In fact, the outputs of *SVD* for a matrix *A* are three matrices *U*, Σ , and V^t . Matrix Σ is a diagonal matrix in which the diameter contains singular values, the number of non-zero values on this diameter determines the rank, non-zero values in this matrix indicate the presence of latent factors in the user-item matrix. But in most cases, among the non-zero

In the table algorithm.

- Data Set: a complete matrix containing all user points to items.
- M_t : Test matrix in which the system test operation is performed.
- *a*: a target user in the test matrix.
- M_a : The resulting reduced matrix obtained from the ratings of the test matrix by eliminating *k* of the user *a* ratings.
- *k*: the number of points given by the target user *a* that has been removed from the test matrix to obtain the M_a matrix.
- $RMSE_k$: an error received by each removed rate.
- $RMSE_a$: an error received by each user *a*.
- $RMSE_t$: The accumulated values $RMSE_a$ are due to the predictions made by the technique used in the problem for each user *a*.

To calculate the accumulated $RMSE_t$ generates an M_a matrix for each user *a*, then, using the Euclidean technique applied to the M_a matrix, the P_i predictions for user *a* is done. Predicted ratings are compared to actual ratings r_i . In this step, the value $RMSE_a$ for the user is calculated using Equation (11).

$$RMSE_a = \sqrt{\frac{1}{n_a} \sum_{i=1}^{n_a} (p_{u,i} - r_{u,i})^2} \quad (11)$$

The value of $RMSE_a$ is then added to $RMSE_t$, and this process is repeated for each user. The accumulated value $RMSE_t$ is calculated for the Euclidean technique as described. Now, after calculating the value of $RMSE_t$, the solution fit is evaluated using the cost function expressed in Equation (12).

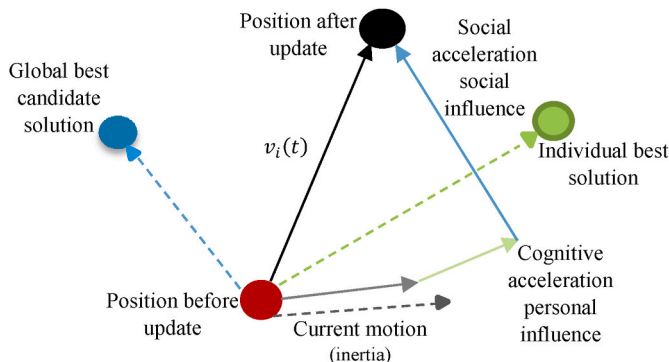


Fig. 1. How particles move in the PSO.

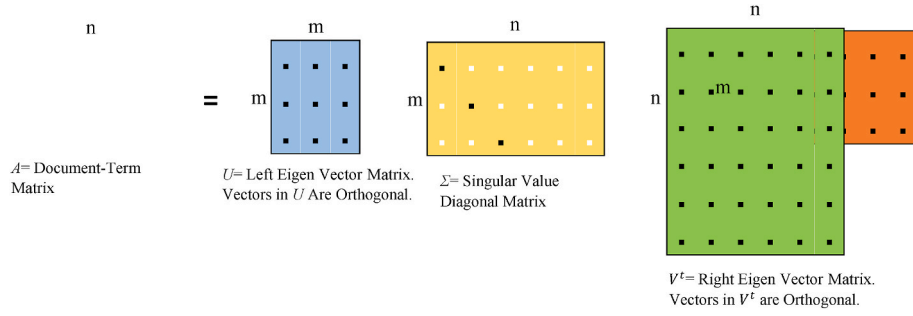


Fig. 2. General method of Singular Value Decomposition.

Fig. 3. Structure of i th particle used by the optimal particle swarm algorithm.Fig. 4. Structure of i th chromosome used by GA.

$$\text{Cost Function} = W_1 \text{RMSE}_t + W_2(1 - q) \quad (12)$$

In the above equation.

$$q = \frac{k}{R - 1} \quad (13)$$

- W_1 and W_2 are weight coefficients to indicate the importance of the technique with values in the range [0,1].
- q is the value calculated by Equation (13), where R is the rating of each user in the test matrix.

The cost function expressed in Equation (12) consists of two components. The first component $W_1 \text{RMSE}_t$ is responsible for summarizing the error in the prediction process. The second component $W_2(1 - q)$ represents the performance of the technique used in the problem. The sum of these components guides the GA through the search space. The whole process described is repeated for each group of algorithms' population to generate a new population, and eventually all of the above processes are repeated for the new population until the cessation conditions are reached (Da Silva et al., 2016).

4.4. Recommendation

In the last phase, after completing the initial hierarchy of models, a list of the best items recommended to the target user is provided.

5. Experimental evaluations

5.1. Data set

In this research, the performance of the proposed method is evaluated using the widely used and standard data sets of Movielens 100k,¹ Movielens 1m² and Filmtrust³ which are in three different sizes. All the mentioned mathematical approaches have been implemented in the data set. The information related to all three data sets is listed in Table 3. It should be noted that all users are randomly selected to log into the system. The scale of ratings for the Movielens data set ranges from 1 to 5 and for the Filmtrust data set ranges from 0.5 to 4.

5.2. Experimental settings

Cross-validation was used to measure the efficiency and effectiveness of our model. The validation method used is cross-validation of the K layer (the value of K is considered equal to 10). Training and testing operations are performed on the data. Each time a portion of the layers is used for system testing, and the $k-1$ layer is used for system training, and this procedure is repeated K times until all the data is used once for the training operation and once for the test operation. The average of the results obtained from this method is considered as a single estimate. Finally, ALLBUT10, ALLBUT5, GIVEN15, and GIVEN10 strategies were used to set up our experiments and calculate the accuracy and error of the proposed method in each of the mentioned strategies.

5.3. Evaluation metrics

We evaluate the proposed hybrid clustering using the *Davies-Bouldin* index defined in Equation (14) and the *Dunn's* index defined in Equation (15) (Barak and Mokfi, 2019). To evaluate our recommender system, we use *RMSE*, *MAE*, *precision*, and *recall* criteria (Ekstrand et al., 2010). Each of the criteria for the target user a is calculated according to equations (16)–(19).

$$DB = \frac{1}{k} \sum_{i=1, i \neq j}^k \max \left(\frac{d(c)_i + d(c)_j}{d(c_i, c_j)} \right) \quad (14)$$

$$D = \frac{\min_{1 \leq i \leq j \leq k} d(c_i, c_j)}{\max_{1 \leq s \leq k} \Delta_s} \quad (15)$$

$$\text{RMSE}_a = \sqrt{\frac{1}{n_a} \sum_{i=1}^{n_a} (p_{a,i} - r_{a,i})^2} \quad (16)$$

$$\text{MAE}_a = \frac{1}{n_a} \sum_{i=1}^{n_a} |p_{a,i} - r_{a,i}| \quad (17)$$

$$\text{Precision} = \frac{|\text{relevant items recommended}|}{|\text{all items retrieved and recommended}|} \quad (18)$$

$$\text{Recall} = \frac{|\text{relevant items recommended}|}{|\text{all relevant items retrieved and not recommended}|} \quad (19)$$

In equation (14), $d(c)_i$ is the measure of scatter within the cluster for cluster c_i which is the average distance of all instances in cluster c_i to their centre c_i , similarly $d(c)_j$ is the measure of scatter within the cluster for cluster c_j , $d(c_i, c_j)$ is a distance of two cluster centers c_i and c_j that is

¹ <https://grouplens.org/datasets/movielens/100k>.

² <https://grouplens.org/datasets/movielens/1m>.

³ <https://guoguibing.github.io/librec/datasets/filmtrust>.

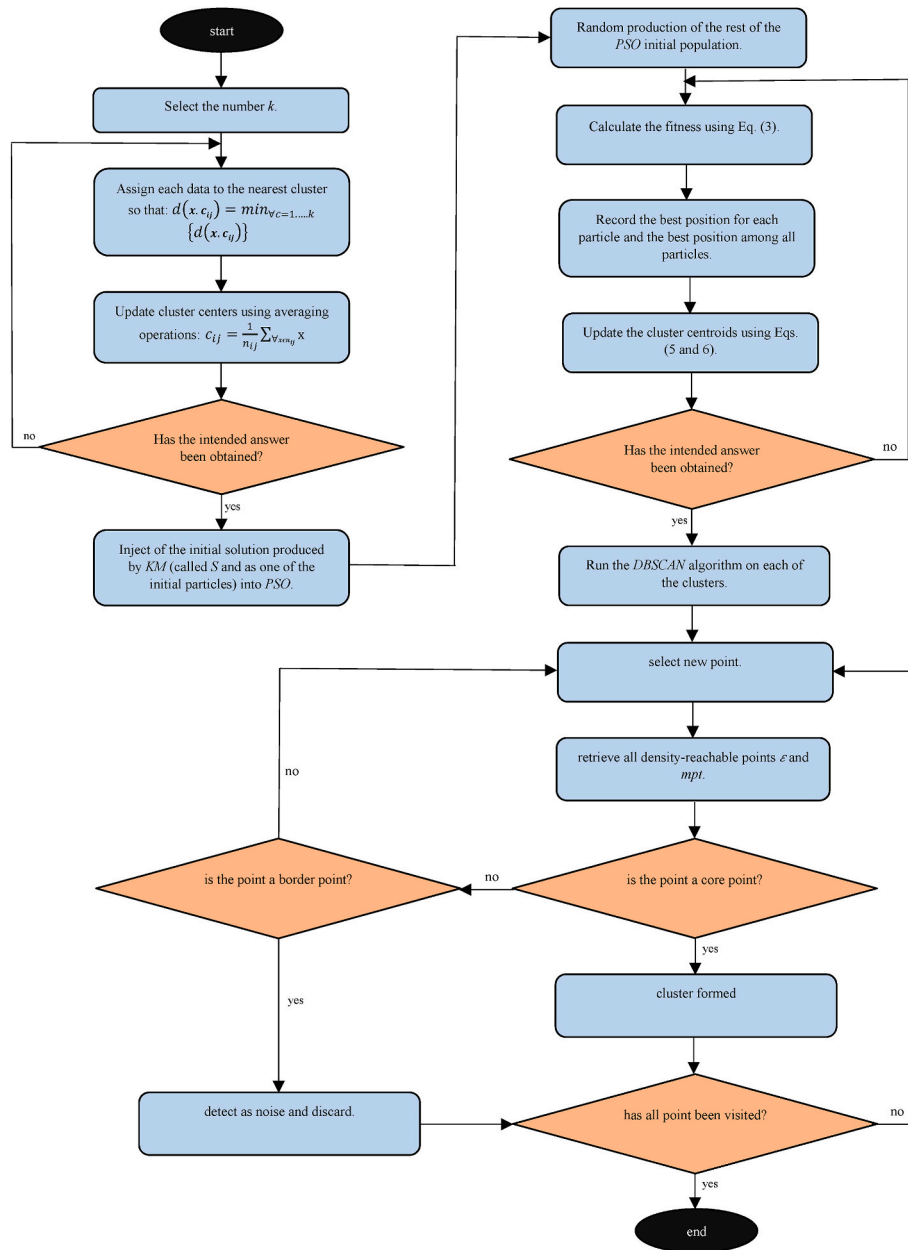


Fig. 5. Flowchart of PSO-KD hybrid algorithm.

called the between-cluster distance, and K is the number of clusters. Since we want to maximize the between-cluster distance and minimize the within-cluster scatter, the aim is to minimize this measure. Also, in equation (15), c_i and c_j are the centers of c_i and c_j clusters, respectively, and Δ_s is the maximum distance between two samples belonging to c_s cluster. In fact, in the numerator of this fraction, the distance between two clusters and in the denominator, the diameter of each cluster can be seen. and the ratio of these two will be a scale to measure the distance between the two clusters. Therefore, the larger the value of *Dunn's* index, the better the separability and, as a result, the more appropriate clustering.

In equations (16) and (17), n is the recommended number of items. $p_{a,i}$ is the rating predicted by the algorithm and $r_{a,i}$ is the actual rating.

5.4. Parameter settings

In this section, experiments were conducted to determine the best values for the parameters of PSO and GA algorithms.

Table 3

Main properties of the data sets used in the experiments.

Data set	Rating	Item	User
Movielens 100k	100000	1682	943
Movielens 1m	1000209	3900	6040
Filmtrust	35497	2071	1508

As we know, C_1 , C_2 and W are PSO algorithm parameters. In order to select the best value for these parameters, according to Tables 4 and 5 levels were determined for each parameter and the algorithm was repeated 125 times with different parameters.

Several samples of the parameters combination, whose results lead to better clustering (minimizing the *Davies-Bouldin* index and maximizing the *Dunn's* index), are listed in Table 5. According to the table, the PSO with parameters $c_1 = 2$, $c_2 = 1.5$ and $W = 0.8$ has obtained the best result.

The parameters considered for PSO are listed in Table 6.

Table 4

Setting of experiment for best parameter combination.

PSO Parameters	Level				
	1	2	3	4	5
C ₁	0.5	1	1.5	2	2.5
C ₂	0.5	1	1.5	2	2.5
W	0.5	0.6	0.7	0.8	0.9

Table 5

Davies-Bouldin and Dunn's index for three datasets with different PSO parameters settings.

PSO Parameters			Movielens 100k		Movielens 1m		Filmtrust	
C ₁	C ₂	W	Davies-Bouldin	Dunn	Davies-Bouldin	Dunn	Davies-Bouldin	Dunn
1	1.5	0.5	3.991	0.0181	4.001	0.0178	4.102	0.0169
1	2	0.5	3.990	0.0184	4.000	0.0179	4.100	0.0170
1.5	2.5	0.7	3.989	0.0188	3.999	0.0180	4.007	0.0171
2	2	0.4	3.984	0.0189	3.997	0.0183	4.005	0.0175
2	1.5	0.8	3.981	0.0190	3.995	0.0185	4.001	0.0178

After determining the PSO algorithm parameters, we determine the GA algorithm parameters. First, we want to show the suitable number of iterations for this algorithm.

In Figs. 6–8, the process of obtaining the best value for the RMSE, Precision, and Recall criteria in the number of different iterations of the GA and in the three available data sets is shown.

According to the results presented in the graphs above, after repeating the GA 100 times, there were no considerable changes in the results, so we are satisfied with this amount of repetition.

After selecting the suitable number of iterations, we determine two other parameters of GA, namely the crossover rate (P_c) and the mutation rate (P_m). Based on Tables 7 and 5 levels are determined for these two parameters.

Several samples of the best parameters combination are listed in Table 8. According to the table, the GA with parameters $P_c = 0.8$ and $P_m = 0.01$ has obtained the best result.

Table 6

The value of the assumed parameters for the PSO.

Parameters	Iteration	Size of population	W	C ₁	C ₂
Values	100	40	0.8	2	1.5

The parameters considered for GA are listed in Tables 9 and 10.

5.5. Experimental results

The experimental plan carried out in this paper consists of three parts. In the first part, the quality of the proposed hybrid clustering method was evaluated. In the experiments of the second part, the quality and superiority of the proposed recommender system were shown compared to the basic steps and other existing methods. In the third part, the statistical significance of the proposed method was analyzed.

5.5.1. Evaluation of the quality of the proposed clustering

In this section, the proposed hybrid clustering method is compared with each of its basic steps using internal evaluation criteria. Therefore, all data sets were clustered from 3 to 8 clusters. The Davies-Bouldin index and Dunn's index were calculated for each clustering output. The results of this measure, in addition to determining the number of suitable clusters, showed the superiority of the combined clustering method over the basic steps. Table 11 contains Davies-Bouldin index values and Table 12 contains Dunn's index values for clustering methods in three different data sets. The results in both tables show that the proposed method has performed better than other methods in the number of

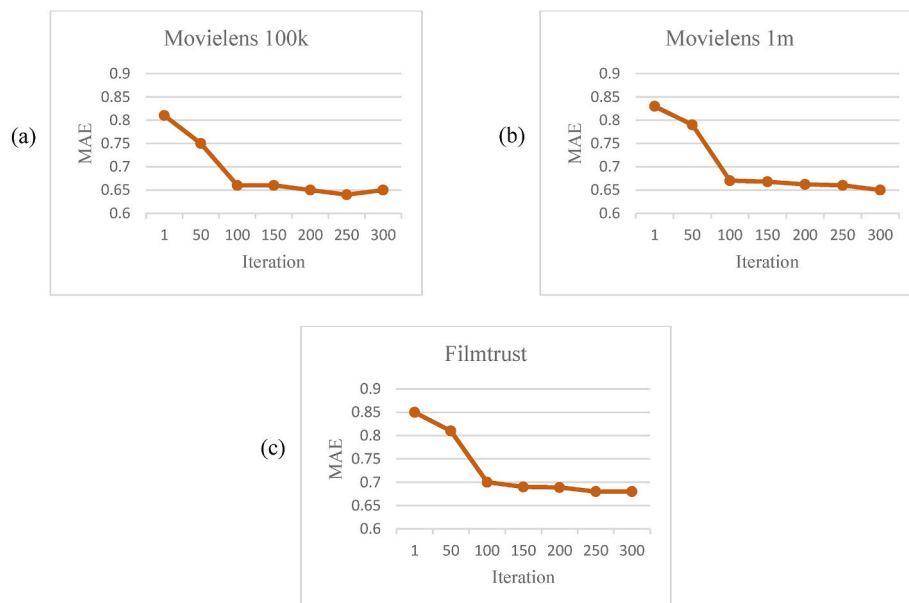


Fig. 6. The process of obtaining the best value for RMSE by GA in the number of different iterations in the three different data sets.

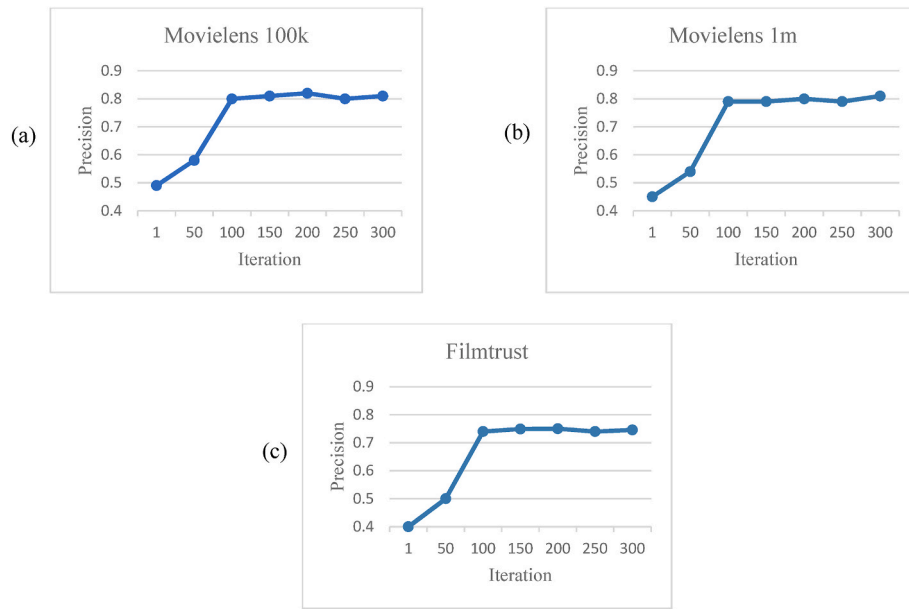


Fig. 7. The process of obtaining the best value for *Precision* by GA in the number of different iterations in the three different data sets.

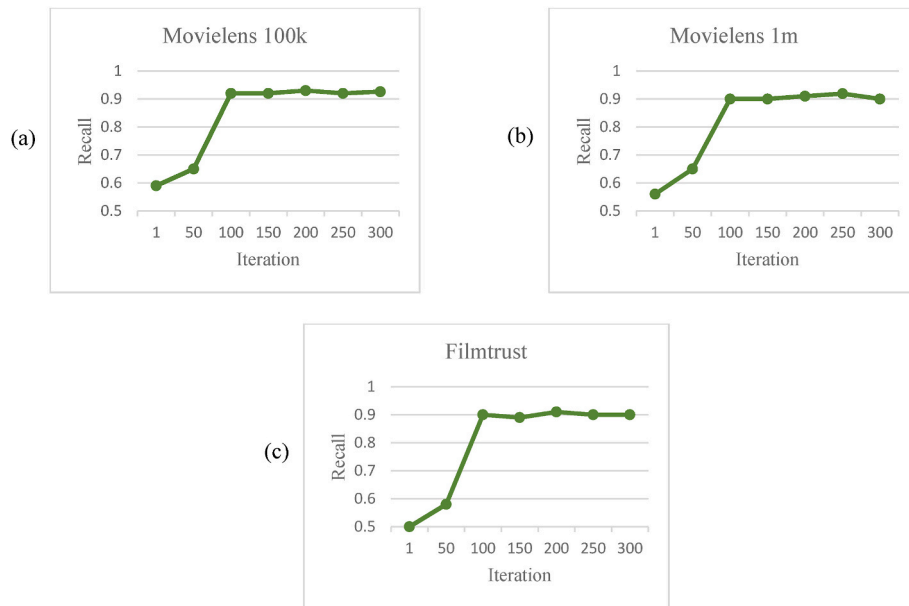


Fig. 8. The process of obtaining the best value for *Recall* by GA in the number of different iterations in the three different data sets.

Table 7
Setting of experiment for best parameter combination.

GA Parameters	Level				
	1	2	3	4	5
P_c	0.8	0.9	0.10	0.11	0.12
P_m	0.01	0.02	0.03	0.04	0.05

different clusters, both in terms of the *Davies-Bouldin* index and *Dunn's* index. In addition, the results showed that all the clustering methods obtained better results in all three data sets when the number of clusters was equal to 5.

5.5.2. Evaluation of the quality of the proposed recommender system

Our comprehensive proposed method consists of *KM* clustering

optimized by the *PSO* algorithm, noise cancellation using *DBSCAN* optimized by GA, reduction of matrix dimension using *SVD* optimized by GA (*KM* + *PSO* + *DBSCAN* + *SVD* + GA).

Therefore, in this section, first, a complete analysis of the system is presented compared to the basic steps used in the method itself. In the next step, the proposed method is compared with other existing methods and the results are displayed.

5.5.2.1. Comparison of the proposed method with the basic steps. In Tables 13–15, we show the improvement of the proposed method compared to the basic steps for *RMSE*, *Precision* and *Recall* criteria. Based on the results presented in Tables 13 and 14, the proposed method has improved in all three data sets and in all four strategies compared to the basic steps in terms of *RMSE* and *Precision* criteria. Based on the results of Table 15, the proposed method has performed better than the basic

Table 8

Precision, Recall and RMSE for three datasets with different GA parameters settings.

Parameters		Movielens 100k			Movielens 1m			Filmtrust		
P_c	P_m	RMSE	Precision	Recall	RMSE	Precision	Recall	RMSE	Precision	Recall
0.9	0.05	0.65	0.74	0.92	0.68	0.73	0.91	0.71	0.69	0.90
0.11	0.02	0.64	0.75	0.92	0.68	0.73	0.91	0.71	0.69	0.90
0.9	0.01	0.64	0.75	0.93	0.68	0.73	0.92	0.69	0.70	0.90
0.8	0.03	0.63	0.75	0.93	0.66	0.74	0.93	0.69	0.70	0.90
0.8	0.01	0.60	0.78	0.94	0.65	0.75	0.93	0.68	0.70	0.91

Table 9

Value of assumed parameters for GA.

Parameters	Iteration	Size of population	P_c	P_m
Values	100	40	0.8	0.01

Table 10

The value of lower and upper bounds for each of the parameters in the proposed method.

Parameters	mpt	ϵ	τ
Values	[5,10]	[10,15]	[0, max_{λ}]

Table 11

Comparison of the results of the proposed hybrid clustering method with other methods using Davies-Bouldin index.

		KM	KM + PSO	DBSCAN	PSO-KD
Data Set	Clusters	Davies-Bouldin			
Movielens 100k	3	5.323	5.201	15.330	4.601
	4	5.901	5.422	15.716	4.101
	5	4.203	4.101	15.100	3.981
	6	4.989	4.111	15.101	4.002
	7	5.777	5.168	17.213	4.602
	8	6.911	6.102	16.701	5.961
Movielens 1m	3	5.701	5.127	14.256	4.950
	4	5.134	5.012	15.901	4.203
	5	5.001	4.218	13.989	3.999
	6	6.106	4.906	15.012	4.112
	7	5.404	5.192	15.961	4.907
	8	7.116	6.220	17.001	5.423
Filmtrust	3	6.301	6.260	17.120	5.206
	4	5.926	5.922	16.931	4.119
	5	5.901	5.804	16.470	4.099
	6	6.292	6.099	18.129	5.003
	7	7.265	6.701	16.909	5.100
	8	7.106	6.912	17.805	5.501

steps in terms of the *Recall* criterion, but only in the Movielens 100k data set and in the Given 10 strategy, the amount of *Recall* in the proposed method and in one of the basic steps is equal.

5.5.2.2. Comparison of the proposed method with other methods. Now the proposed method is compared with the some available methods such as UPCC, PCA-k-means, PCA-SOM, PCA-GAKM and ABC-KM (Z. Wang et al., 2014; Katarya, 2018) and the results are shown. The results in Table 16 showed that our proposed method is better than other methods in terms of error. Next, in Tables 17–19, the *precision*, *recall*, and *MAE* of the proposed method with other available methods and in different numbers of clusters are reviewed. The results in three different data sets showed that the proposed method provided the best results for any number of clusters, both in terms of error and accuracy compared to other methods. Also, the results showed that the proposed method performed better in the Movielens 100k data set than the other two data sets, except for a few cases (in Tables 17 and in $k = 5$, the Movielens 100k and Movielens 1m data sets showed the same results and in $k = 20$ Movielens 1m data set has performed better, in Table 18 only in $k = 4$

Table 12

Comparison of the results of the proposed hybrid clustering method with other methods using Dunn's index.

		KM	KM + PSO	DBSCAN	PSO-KD
Data Set	Clusters	Dunn			
Movielens 100k	3	0.0146	0.0151	0.0130	0.0174
	4	0.0109	0.0120	0.0101	0.0140
	5	0.0180	0.0172	0.0160	0.0199
	6	0.0140	0.0149	0.0128	0.0161
	7	0.0142	0.0150	0.0130	0.0165
	8	0.0120	0.0128	0.0100	0.0136
Movielens 1m	3	0.0140	0.0148	0.0129	0.0158
	4	0.0104	0.0115	0.0100	0.0131
	5	0.0168	0.0175	0.0140	0.0188
	6	0.0139	0.0146	0.0131	0.0155
	7	0.0148	0.0155	0.0133	0.0162
	8	0.0103	0.0110	0.0100	0.0126
Filmtrust	3	0.0134	0.0145	0.0121	0.0152
	4	0.0106	0.0111	0.0101	0.0120
	5	0.0159	0.0167	0.0131	0.0178
	6	0.0135	0.0141	0.0124	0.0150
	7	0.0143	0.0154	0.0128	0.0159
	8	0.0105	0.0119	0.0102	0.0130

Table 13

RMSE values for the proposed method and basic steps in three different data sets.

Data set	System/Strategy	Given 15	Given 10	Allbut 5	Allbut 10	Average
Movielens 100k	KM	0.69	0.79	0.67	0.70	0.71
	KM + DBSCAN	0.60	0.80	0.75	0.70	0.71
	KM + PSO + DBSCAN	0.60	0.80	0.60	0.69	0.67
	Proposed method	0.57	0.70	0.50	0.64	0.60
Movielens 1m	KM	0.74	0.87	0.67	0.68	0.74
	KM + DBSCAN	0.70	0.88	0.76	0.70	0.76
	KM + PSO + DBSCAN	0.75	0.79	0.69	0.70	0.73
	Proposed method	0.68	0.70	0.65	0.65	0.67
Filmtrust	KM	0.75	0.83	0.70	0.81	0.77
	KM + DBSCAN	0.70	0.98	0.76	0.77	0.80
	KM + PSO + DBSCAN	0.68	0.85	0.60	0.73	0.71
	Proposed method	0.64	0.78	0.55	0.68	0.66

Movielens 100k and Movielens 1m data sets showed the same results, in Table 19 in $k = 6$ Movielens 100k data sets and Filmtrust showed the same results).

Finally, in Table 20, to show the success of the proposed method in improving the non-scalability problem, the proposed method was compared with other methods in terms of recommendation speed. The execution time of the proposed method in the first, second and third data set is 25.89, 37.24 and 29.98 s, respectively. The results show that our method performs the recommendation in less time than other methods,

Table 14

Precision values for the proposed method and basic steps in three different data sets.

Data set	System/ Strategy	Given 15	Given 10	Allbut 5	Allbut 10	Average
Movielens 100k	KM	0.60	0.45	0.70	0.61	0.59
	KM + DBSCAN	0.55	0.36	0.60	0.58	0.52
	KM + PSO + DBSCAN	0.63	0.49	0.70	0.64	0.61
	Proposed method	0.73	0.54	0.78	0.74	0.69
Movielens 1m	KM	0.59	0.40	0.65	0.68	0.58
	KM + DBSCAN	0.50	0.38	0.59	0.54	0.50
	KM + PSO + DBSCAN	0.60	0.49	0.69	0.62	0.60
	Proposed method	0.69	0.52	0.75	0.71	0.66
Filmtrust	KM	0.50	0.25	0.60	0.55	0.47
	KM + DBSCAN	0.45	0.25	0.51	0.50	0.42
	KM + PSO + DBSCAN	0.54	0.31	0.60	0.58	0.50
	Proposed method	0.65	0.45	0.70	0.65	0.61

Table 15

Recall values for the proposed method and basic steps in three different data sets.

Data set	System/ Strategy	Given 15	Given 10	Allbut 5	Allbut 10	Average
Movielens 100k	KM	0.90	0.85	0.89	0.89	0.88
	KM + DBSCAN	0.89	0.80	0.85	0.80	0.83
	KM + PSO + DBSCAN	0.91	0.91	0.90	0.91	0.90
	Proposed method	0.93	0.91	0.95	0.93	0.93
Movielens 1m	KM	0.87	0.80	0.88	0.87	0.85
	KM + DBSCAN	0.80	0.80	0.84	0.81	0.81
	KM + PSO + DBSCAN	0.87	0.87	0.90	0.88	0.88
	Proposed method	0.91	0.89	0.93	0.90	0.90
Filmtrust	KM	0.85	0.80	0.85	0.86	0.84
	KM + DBSCAN	0.80	0.78	0.81	0.81	0.80
	KM + PSO + DBSCAN	0.86	0.85	0.89	0.91	0.87
	Proposed method	0.90	0.89	0.92	0.90	0.90

except for one case which belongs to the k-means cluster method in the Movielens 100k data set. However, with the increase in the size of the data set, in addition to other methods, the efficiency of our method increases in the k-means cluster method. Therefore, considering the good results that the proposed method has achieved in terms of error and accuracy, this slight difference in run time of a data set can be ignored.

5.5.3. Examining the proposed recommender system using statistical test

In this section, the performance of the proposed method and its basic steps were statistically examined. As shown in Tables 13–15, our proposed recommender system has performed better than its basic steps in terms of RMSE, Precision, and Recall. Now, in order to test whether the difference in error rate and accuracy between the proposed method and the basic steps was statistically significant, a *t*-test and a Wilcoxon signed rank test (Li et al., 2010) were performed.

Two hypotheses were used to perform this test. The primary hypothesis was called the null hypothesis, which was denoted by H_0 and showed that the variable levels in the two groups did not differ from

Table 16

Comparison of the MAE between the proposed method and different approaches.

Data set	System/ Strategy	Given 15	Given 10	Allbut 5	Allbut 10
Movielens 100k	UPCC	0.84	0.90	0.79	0.80
	PCA-SOM	0.79	0.82	0.76	0.79
	PCA-GAKM	0.76	0.80	0.75	0.78
	ABC-KM	0.77	0.78	0.74	0.76
	Proposed method	0.55	0.69	0.43	0.56
Movielens 1m	UPCC	0.86	0.90	0.81	0.81
	PCA-SOM	0.80	0.85	0.78	0.76
	PCA-GAKM	0.80	0.80	0.78	0.78
	ABC-KM	0.78	0.85	0.80	0.75
	Proposed method	0.59	0.75	0.49	0.62
Filmtrust	UPCC	0.88	0.93	0.85	0.84
	PCA-SOM	0.85	0.90	0.82	0.80
	PCA-GAKM	0.82	0.81	0.80	0.72
	ABC-KM	0.85	0.91	0.75	0.72
	Proposed method	0.65	0.77	0.58	0.70

Table 17

Comparison MAE between the proposed method and different approaches in different values of *k*.

Data set	System/ cluster	5	10	15	20	25	30
Movielens 100k	PCA-GAKM	0.79	0.77	0.77	0.78	0.78	0.78
	PCA-SOM	0.82	0.79	0.79	0.79	0.80	0.80
	UPCC	0.82	0.82	0.82	0.82	0.82	0.82
	k-means cluster	0.82	0.82	0.81	0.81	0.81	0.81
	ABC-KM	0.77	0.76	0.76	0.77	0.78	0.78
Movielens 1m	Proposed method	0.66	0.65	0.67	0.70	0.66	0.69
	PCA-GAKM	0.81	0.80	0.79	0.80	0.78	0.80
	PCA-SOM	0.85	0.78	0.79	0.84	0.81	0.82
	UPCC	0.85	0.85	0.85	0.85	0.83	0.85
	k-means cluster	0.86	0.82	0.84	0.83	0.85	0.82
Filmtrust	ABC-KM	0.78	0.70	0.78	0.79	0.80	0.81
	Proposed method	0.66	0.66	0.68	0.69	0.68	0.70
	PCA-GAKM	0.85	0.84	0.87	0.82	0.81	0.80
	PCA-SOM	0.88	0.86	0.90	0.88	0.86	0.89
	UPCC	0.90	0.90	0.92	0.90	0.88	0.85
	k-means cluster	0.91	0.86	0.88	0.90	0.93	0.87
	ABC-KM	0.81	0.80	0.84	0.80	0.80	0.85
	Proposed method	0.70	0.71	0.74	0.73	0.69	0.75

each other. There was also an alternative hypothesis H_1 , which competed with the null hypothesis and showed that the amount of the variable in the two groups was different from each other.

The probability of error of $\alpha = 0.05$ was used in this experiment to determine the test's significance level. After performing the statistical test, deciding whether the test was significant or not depended on the calculated *p*-value. In fact, if the *p*-value obtained from each test was greater than the value of α (*p*-value > 0.05), the null hypothesis was confirmed and the test was not significant. If the value of *p* was less than or equal to α (*p*-value ≤ 0.05), the null hypothesis was rejected and the test was significant.

Table 21 showed the results of the *p*-value of the statistical test. The obtained results indicated that the error and accuracy difference between the proposed method and the basic steps was statistically significant (in all steps, the *p*-value was less than 0.05).

Table 18Comparison *Precision* between the proposed method and different approaches in different values of *k*.

Data set	System/cluster	3	4	5	6	7	8
Movielens 100k	<i>PCA-GAKM</i>	0.40	0.39	0.38	0.37	0.37	0.36
	<i>PCA-SOM</i>	0.38	0.39	0.38	0.36	0.35	0.34
	<i>UPCC</i>	0.31	0.30	0.29	0.29	0.28	0.28
	<i>k-means cluster</i>	0.11	0.14	0.12	0.11	0.12	0.11
	<i>ABC-KM</i>	0.42	0.41	0.40	0.41	0.40	0.40
	<i>Proposed method</i>	0.79	0.77	0.80	0.79	0.78	0.71
Movielens 1m	<i>PCA-GAKM</i>	0.39	0.35	0.38	0.35	0.34	0.35
	<i>PCA-SOM</i>	0.35	0.34	0.35	0.30	0.32	0.30
	<i>UPCC</i>	0.28	0.25	0.29	0.27	0.25	0.22
	<i>k-means cluster</i>	0.15	0.12	0.10	0.11	0.10	0.10
	<i>ABC-KM</i>	0.42	0.39	0.40	0.38	0.36	0.35
	<i>Proposed method</i>	0.75	0.77	0.79	0.70	0.75	0.70
Filmtrust	<i>PCA-GAKM</i>	0.35	0.30	0.37	0.30	0.32	0.30
	<i>PCA-SOM</i>	0.30	0.30	0.32	0.28	0.29	0.25
	<i>UPCC</i>	0.25	0.21	0.27	0.22	0.24	0.20
	<i>k-means cluster</i>	0.08	0.12	0.09	0.11	0.11	0.10
	<i>ABC-KM</i>	0.40	0.35	0.39	0.35	0.36	0.30
	<i>Proposed method</i>	0.69	0.70	0.74	0.70	0.69	0.70

Table 19Comparison *Recall* between the proposed method and different approaches in different values of *k*.

	System/cluster	3	4	5	6	7	8
Movielens 100k	<i>PCA-GAKM</i>	0.11	0.13	0.20	0.30	0.40	0.50
	<i>PCA-SOM</i>	0.11	0.13	0.25	0.30	0.32	0.40
	<i>UPCC</i>	0.10	0.13	0.18	0.20	0.30	0.40
	<i>k-means cluster</i>	0.04	0.05	0.06	0.07	0.08	0.08
	<i>ABC-KM</i>	0.20	0.25	0.29	0.35	0.39	0.44
	<i>Proposed method</i>	0.89	0.89	0.92	0.89	0.89	0.90
Movielens 1m	<i>PCA-GAKM</i>	0.11	0.12	0.18	0.25	0.38	0.45
	<i>PCA-SOM</i>	0.10	0.11	0.20	0.28	0.30	0.35
	<i>UPCC</i>	0.10	0.12	0.15	0.19	0.25	0.38
	<i>k-means cluster</i>	0.04	0.05	0.06	0.07	0.08	0.08
	<i>ABC-KM</i>	0.18	0.23	0.25	0.33	0.39	0.40
	<i>Proposed method</i>	0.85	0.85	0.90	0.86	0.80	0.85
Filmtrust	<i>PCA-GAKM</i>	0.10	0.10	0.15	0.20	0.35	0.44
	<i>PCA-SOM</i>	0.11	0.10	0.18	0.25	0.30	0.31
	<i>UPCC</i>	0.10	0.11	0.12	0.15	0.21	0.35
	<i>k-means cluster</i>	0.03	0.05	0.04	0.04	0.05	0.03
	<i>ABC-KM</i>	0.15	0.20	0.22	0.29	0.35	0.37
	<i>Proposed method</i>	0.79	0.84	0.90	0.89	0.79	0.85

6. Conclusion

Due to the vast amount of unstructured data on the internet, recommender systems are still a fascinating area for research. These systems have identified similar users by checking users and according to their preferences, remove information barriers and provide useful and targeted recommendations.

An important part of the success or failure of e-businesses is related to the strength of their recommendation engines. Considering the importance of recommender systems, in this article, a hybrid method using optimization algorithms for clustering and data dimension reduction in the offline phase of recommender systems based on collaborative filtering in order to improve the quality of the system, reduction the execution time and solve the problem of lack of scalability is presented.

To perform the clustering, the combined *PSO-KD* algorithm (the result of combining *KM* clustering optimized by *PSO* and *DBSCAN* clustering optimized by *GA*) has been used. Using this combined method

will cover the disadvantages of the *KM* algorithm and will be very effective in clustering large data. In the following, the *SVD* method optimized by *GA* is used to matrix dimension reduction.

For evaluate the quality of the proposed hybrid clustering, the clustering performed using two internal evaluation criteria including *Davies-Bouldin* index and *Dunn* index was evaluated on three data set with different dimensions named Movielens 100k, Movielens 1m and Filmtrust. The results show the proposed hybrid clustering has obtained better results.

Also, on the other side, the proposed recommender system was evaluated on each of the strategies ALLBUT10, ALLBUT5, GIVEN15, GIVEN10 and in the existing data set. Experiments show that the proposed method has been able to reduce the *RMSE* and *MAE* in terms of Rating prediction for the items and also has been able to increase the *Precision* and *Recall* in terms of items recommendation to the target user. On the other side, the results show that the use of clustering and dimension reduction methods has a positive effect on the execution time.

Table 20

Comparison of execution time (recommendation per second) between the proposed method and different approaches.

Data sets/Method	<i>PCA-GAKM</i>	<i>PCA-SOM</i>	<i>UPCC</i>	<i>k-means cluster</i>	<i>PCA-k-means</i>	<i>ABC-KM</i>	<i>Proposed method</i>
Movielens 100k	26.32	141.73	159.34	16.25	55.56	53.32	25.89
Movielens 1m	43.61	159.11	171.04	45.99	67.54	64.12	37.24
Filmtrust	34.79	150.48	164.91	37.15	59.22	56.11	29.98

Table 11

P-value results using statistical tests to compare *RMSE*, *Precision* and *Recall* between the proposed method and each of the basic steps.

	Movielens 100k			Movielens 1m			Filmtrust		
Measure	RMSE	Precision	Recall	RMSE	Precision	Recall	RMSE	Precision	Recall
Comparison between proposed method and <i>KM</i>									
<i>t</i> -test	0.00016	0.00033	0.00020	0.00009	0.00026	0.00017	0.00007	0.00027	0.00019
wilcoxon	0.00005	0.00011	0.00011	0.00014	0.00111	0.00015	0.00001	0.00012	0.00009
Comparison between proposed method and <i>KM</i> + <i>DBSCAN</i>									
<i>t</i> -test	0.00218	0.00021	0.00019	0.00022	0.00015	0.00031	0.00017	0.00035	0.00030
wilcoxon	0.00003	0.00009	0.00007	0.00005	0.00011	0.00011	0.00004	0.00011	0.00015
Comparison between proposed method and <i>KM</i> + <i>PSO</i> + <i>DBSCAN</i>									
<i>t</i> -test	0.00015	0.00221	0.00022	0.00111	0.00108	0.00212	0.00013	0.00007	0.00018
wilcoxon	0.00001	0.00007	0.00019	0.00009	0.00015	0.00022	0.00005	0.00009	0.00011

CRediT authorship contribution statement

Zahra Movafegh: Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft. **Abdoreza Rezapour:** Conceptualization, Methodology, Validation, Resources, Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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